

Wu Caiyan

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The Chinese University of Hong Kong,
Shenzhen
2001 Longxiang Road, Longgang
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EDUCATION

The Chinese University of Hong Kong, Shenzhen

Expected May 2028

- School of Data Science*: B.E. in Computer Science and Engineering
- Cumulative GPA*: 3.69/4.00
- Honors*: Dean's List (2024–2025)
- Core Courses*: Data Structures and Advanced Programming, Discrete Mathematics, Introduction to Computer Systems, Machine Learning, Optimization, Probability and Statistics

RELATED EXPERIENCE

School of Science and Engineering, Undergraduate Student Teaching Fellow, Shenzhen, China January 2026 - Present

- Facilitate weekly tutorial sessions for 31 students, designing supplementary practice problems that help reduce common errors in assignments
- Evaluate over 30 assignments and deliver detailed feedback, enabling students to strengthen problem-solving accuracy

Tencent Mini Goose Offline Tech Innovation Camp, Camp Course Assistant, Chengdu, China July 2025 - August 2025

- Assisted technical instructors with code debugging and Q&A for 18 students, resolving issues in front-end layout, component invocation, and basic syntax
- Guided 4 student teams through the full process from topic selection to functional development; facilitated demonstrable AI mini-program projects that passed the final review

Study Programme: Yangtze River Delta and Pearl River Delta, Research Student, Shenzhen, China July 2025

- Achieved selection in a joint research program by CUHK-Shenzhen, CUHK, and Ningbo University; investigated port-city symbiosis patterns across four cities (Hangzhou, Ningbo, Shenzhen and Hong Kong) over two weeks, collecting primary data from technology companies including Tencent and Forms Syntron
- Led a group research project on “Port and Urban Symbiosis: Past and Present”, integrating public economic data and corporate interview records to build a multi-dimensional comparative analysis framework; demonstrated findings at the program's closing ceremony

PROJECT EXPERIENCE

RISC-V Instruction Scheduler, Developer, Shenzhen, China February 2026 - March 2026

- Implemented a RISC-V instruction scheduler based on a simplified LLVM model using C++, covering dependency graph construction, priority calculation, and full List Scheduling workflow
- Built an instruction-level directed acyclic graph (DAG) to detect three types of data dependencies (RAW/WAR/WAW), optimized to complete full dependency analysis in a single pass
- Developed List Scheduling algorithm with an Inflight queue, distinguished release timing of true dependencies and order dependencies to reduce pipeline stalls

Python-Based Multi-Line Console Text Editor, Developer, Shenzhen, China February 2025 - May 2025

- Designed and implemented a Vim-style multi-line console text editor supporting over 20 editing commands (insert, delete, cursor movement, line copy/paste, etc.)
- Devised top-down modular architecture with dictionary-based command mapping; integrated stack data structures for multi-level state tracking, demonstrating proficiency in Python program design and data structure application
- Configured regular expressions for word-level cursor jump logic; achieved dual-layer visual display (inline cursor highlighting and line-level marking) via ANSI escape sequences

HONORS AND AWARDS

- Guangdong Contemporary Undergraduate Mathematical Contest in Modeling, Second Prize (2025)
- Outstanding Student Leader Award (2025)
- Shaw Service Award, Shaw College (2025)
- Men's Soccer Competition at the Eighth CUHK-Shenzhen Freshmen Cup, Gold Medal (2024)

SKILLS AND INTERESTS

Programming Languages: Python, C++, MATLAB

Interests: Piano (Amateur Level 10), Football (Champion of Freshman Football Cup 2024), Calligraphy (Chinese Hard-pen & Brush Calligraphy, Level 10)